NEO-TEEM

A Unified Framework of Gravity, Time, and Cosmic Expansion

NS(2026 May)

Abstract

We present **NEO-TEEM**, a unified physical framework in which gravity, time, entropy, cosmic expansion, black-hole behavior, and light propagation all emerge from the dynamics of three fundamental energy states: vibrational energy (v), condensed energy (m), and spatial degrees of freedom (R). The central organizing principle of the theory is the **information density** R/m, which determines the local rate of vibrational evolution and whose spatial gradients generate gravitational attraction. In this view, **time** is the rate of vibrational change, **entropy** is the growth of R, and **gravity** is the tendency of systems to move toward regions of lower information density.

By tracking the evolution of the three energy states, we derive a cosmological expansion equation in which **cosmic acceleration arises from the irreversible increase of R**, eliminating the need for dark energy. The effective gravitational influence of matter increases through the conversion $m \to R$, reproducing the observational role of dark matter without invoking new particles. The resulting expansion history matches ΛCDM-like behavior while offering a simpler physical interpretation.

Black holes appear as **finite information-density structures** where R/m approaches its minimum value, avoiding singularities and providing a natural explanation for horizon behavior, entropy scaling, and Hawking-like radiation as slow reconfiguration of R. Light propagation—whether in optical media or gravitational fields—follows from the same principle: **refraction and lensing both arise from spatial variations in R**, unifying optical and gravitational bending.

NEO-TEEM therefore offers a coherent and physically grounded framework that reproduces the empirical successes of General Relativity and ΛCDM while providing new insights and testable predictions. This paper establishes the foundations of the theory; companion works will present detailed cosmological fits and a full treatment of black-hole and lensing phenomena.

1. Introduction

Modern physics describes the universe through two powerful but fundamentally separate frameworks. General Relativity explains gravity as the curvature of spacetime, while quantum theory governs the behavior of matter and radiation. Cosmology, built on these foundations, supplements the gravitational sector with two hypothetical components—dark matter and dark energy—to match observations of structure formation and accelerated expansion. Despite their empirical success, these frameworks leave several conceptual gaps unresolved:

- the physical origin of gravity,

- the nature of cosmic acceleration,
- the meaning of entropy and information in spacetime,
- the internal structure of black holes,
- and the absence of direct evidence for dark matter or dark energy.

This paper introduces NEO-TEEM, a new physical framework designed to address these gaps through a single unifying principle: the dynamics of information density.

NEO-TEEM is built on three fundamental energy states:

- **v** — vibrational energy (radiation, fluctuations),
- **m** — condensed energy (matter, structure),
- **R** — spatial degrees of freedom (information capacity).

These states interact through a natural hierarchy of energy flow,

$$v \to m \to R, v \to R,$$

which expresses the tendency of the universe to increase its available information capacity. In this view, **entropy is not an auxiliary thermodynamic quantity but a primary physical degree of freedom represented by R.**

A central concept of NEO-TEEM is the **information density**, defined as the ratio

$$I = \frac{R}{m}.$$

This ratio determines how much spatial freedom exists per unit of condensed energy. We show that **gravity emerges as the gradient of information density**, leading to a gravitational potential of the form

$$\Phi = \frac{c^2}{\epsilon}\left(\frac{R}{m} - 1\right),$$

and a corresponding acceleration

$$g = -\nabla\Phi.$$

This formulation reproduces Newtonian gravity when $R(r)$decreases as $1/r$, and it provides a direct physical interpretation of gravitational attraction as a flow toward regions of lower information density.

The same information-density framework naturally extends to cosmology. By tracking the evolution of the three energy states, we derive a Friedmann-like expansion equation in which **cosmic acceleration arises from the growth of R**, without invoking dark energy. Similarly, the effective gravitational influence of matter increases through the conversion $m \to R$, providing a natural explanation for phenomena typically attributed to dark matter.

NEO-TEEM therefore offers a unified description of:

- gravity as information-density gradients,
- time as the local rate of vibrational evolution,
- entropy as the growth of spatial degrees of freedom,
- cosmic expansion without dark components,
- and black holes as finite information-density structures.

The purpose of this paper is to establish the foundations of NEO-TEEM, derive its core equations, and demonstrate its consistency with key cosmological observations. Companion papers will address detailed applications to black holes, gravitational lensing, and the optical properties of matter.

2. The Three-State Energy Framework

The NEO-TEEM framework is built on the premise that all physical phenomena can be described through the interaction of **three fundamental energy states**. These states—vibrational energy (v), condensed energy (m), and spatial degrees of freedom (R)—form the minimal set required to describe the dynamics of matter, radiation, gravity, entropy, and cosmic expansion within a single unified structure.

Rather than treating spacetime, matter, and radiation as separate ontological entities, NEO-TEEM views them as **different organizational modes of a single underlying energy substrate**. This section defines the three states and the transitions between them.

2.1 Vibrational Energy (v)

The **v-state** represents energy stored in oscillatory or fluctuating modes:

- electromagnetic radiation,
- thermal motion,
- quantum fluctuations,
- any form of delocalized or wave-like excitation.

Its defining characteristics are:

- **high mobility**,
- **low information confinement**,
- **rapid temporal evolution**,
- **minimal contribution to gravitational information density**.

In NEO-TEEM, v determines the **local rate of physical evolution**—effectively the "clock speed" of a region. Regions with higher information density (lower R/m) suppress vibrational evolution, producing gravitational time dilation.

2.2 Condensed Energy (m)

The **m-state** corresponds to energy stored in localized, structured, or bound configurations:

- atoms and molecules,
- condensed matter,
- baryonic mass,
- any form of rest-mass energy.

Its defining properties are:

- **high information confinement**,
- **low mobility**,
- **ability to shape the surrounding information density**,

- **dominant contributor to gravitational structure**.

In NEO-TEEM, m is not merely "mass" but a **measure of how strongly energy is localized**, reducing the available spatial degrees of freedom in its vicinity.

2.3 Spatial Degrees of Freedom (R)

The **R-state** represents the **information capacity of space itself**—the number of available microstates or degrees of freedom per unit volume.

R is not geometry; it is **the substrate that geometry emerges from**.

Key properties:

- **R increases irreversibly**, representing entropy growth.
- **R determines the "softness" or "openness" of space**.
- **R controls how easily vibrational energy can propagate**.
- **R is the driver of cosmic expansion**.

In this framework, the expansion of the universe is interpreted as the **global increase of R**, not as the stretching of a geometric manifold.

2.4 Energy Flow: $v \to m \to R$

The three states are connected through a natural hierarchy of transitions:

$$v \to m \to R, v \to R.$$

This expresses the universal tendency for energy to move toward **greater information capacity**.

- **v → m**: Vibrational energy condenses into structured forms (cooling, binding, collapse).
- **m → R**: Localized structures eventually dissolve or thermalize, releasing their confined information into the spatial degrees of freedom.
- **v → R**: Radiation redshifts and loses energy directly to the expansion of R.

This hierarchy provides a physical basis for:

- entropy increase,
- cosmic expansion,
- the arrow of time,
- the emergence of structure,
- the decay of structure,
- and the long-term evolution of the universe.

2.5 Entropy as the Growth of R

In NEO-TEEM, entropy is not an emergent statistical quantity but a **fundamental physical degree of freedom** represented by R.

- When R increases, the universe gains new microstates.
- When m decreases (through decay or thermalization), its information is transferred to R.
- When v redshifts, its energy contributes to R.

Thus:

$$\Delta S \propto \Delta R.$$

This identification allows entropy, gravity, and cosmic expansion to be described within a single unified language.

2.6 Summary of the Three-State Framework

The three states can be summarized as follows:

State	Physical Meaning	Mobility	Information Confinement	Role in Gravity	Role in Expansion
v	Vibrational energy	High	Low	Weak	Redshifts into R
m	Condensed energy	Low	High	Strong	Converts to R
R	Spatial degrees of freedom	N/A	N/A	Defines information density	Drives expansion

Together, these states form the foundation of NEO-TEEM. The next section introduces the central concept that links them: **information density**, the ratio R/m, which determines gravitational behavior.

3. Information Density as the Origin of Gravity

Gravity in NEO-TEEM does not arise from geometric curvature or from a fundamental force. Instead, it emerges from **gradients in information density**, a quantity that measures how much spatial freedom exists per unit of condensed energy.

This section introduces the definition of information density, derives the gravitational potential from it, and shows how Newtonian gravity naturally follows from simple assumptions about the spatial distribution of R.

3.1 Definition of Information Density

In NEO-TEEM, the fundamental quantity governing gravitational behavior is the **information density**, defined as

$$I = \frac{R}{m},$$

where:

- **R** is the spatial information capacity (degrees of freedom),
- **m** is condensed energy (localized structure).

The ratio R/mexpresses **how much "room" space has to reorganize itself per unit of localized energy**. Regions with low $R/m$are "information-dense" and resist vibrational evolution, while regions with high R/mallow rapid evolution.

This simple ratio replaces geometric curvature as the driver of gravitational phenomena.

3.2 Physical Interpretation

Information density determines:

- how easily vibrational energy propagates,
- how fast local clocks run,
- how strongly matter influences its surroundings,
- and how space responds to the presence of energy.

A lower value of R/m corresponds to:

- **slower time**,
- **higher gravitational potential**,
- **greater confinement of vibrational modes**,
- **stronger attraction toward that region**.

Thus, gravity is not a force but a **tendency of systems to move toward regions of lower information density**, where vibrational evolution is suppressed.

3.3 Gravitational Potential from Information Density

The gravitational potential in NEO-TEEM is defined as

$$\Phi = \frac{c^2}{\epsilon}\left(\frac{R}{m} - 1\right),$$

where ϵ is a constant setting the scale of information-density response.

Key features:

- Φ decreases when R/m decreases.
- $\Phi = 0$ corresponds to free space with $R = m$.
- Gravitational wells correspond to regions where $R/m < 1$.

This potential is not postulated; it follows from the requirement that:

- time dilation depends on R/m,
- vibrational evolution slows in information-dense regions,
- and energy flows toward lower information density.

3.4 Gravitational Acceleration

The gravitational acceleration is obtained by taking the spatial gradient of the potential:

$$g = -\nabla\Phi = -\frac{c^2}{\epsilon}\nabla\left(\frac{R}{m}\right).$$

Thus:

Gravity is the spatial gradient of information density.

This formulation is conceptually simple and physically transparent:

- No curvature is required.
- No force field is required.

- No dark matter is required.
- Only the distribution of R and m matters.

3.5 Recovering Newtonian Gravity

To recover the familiar inverse-square law, we assume that the presence of a mass M reduces the surrounding information capacity R in proportion to $1/r$:

$$R(r) = R_\infty - \frac{kM}{r}.$$

Substituting into the information density:

$$\frac{R(r)}{m} = \frac{R_\infty}{m} - \frac{kM}{mr}.$$

Taking the gradient:

$$g(r) = -\nabla\Phi \propto -\nabla\left(\frac{1}{r}\right) = -\frac{1}{r^2}.$$

Thus, the inverse-square law emerges directly from the spatial profile of R.
This derivation shows that:

- **Newtonian gravity is a special case of information-density gradients**,
- not a fundamental force,
- and not a geometric effect.

3.6 Comparison with General Relativity

General Relativity interprets gravity as curvature of spacetime. NEO-TEEM interprets gravity as **variations in information capacity**.
The two frameworks agree in weak fields because:

- curvature reduces the number of available microstates,
- which corresponds to a decrease in R,
- which lowers R/m,
- which produces gravitational attraction.

Thus, GR's curvature is the geometric manifestation of an underlying information-density gradient.
NEO-TEEM provides the physical mechanism behind the geometry.

3.7 Summary

- Information density $R/m$is the fundamental quantity governing gravitational behavior.
- Gravitational potential is proportional to $R/m - 1$.
- Gravitational acceleration is the gradient of information density.
- Newtonian gravity emerges naturally from a $1/r$profile of R.
- GR is recovered as the geometric limit of information-density variation.

This establishes gravity as an emergent phenomenon rooted in the distribution and flow of information in space.

The next section extends this framework to the nature of time and dynamical evolution.

4. Time and Dynamics in NEO-TEEM

Time in NEO-TEEM is not a geometric coordinate, nor an external parameter imposed on physical systems. Instead, **time is the local rate at which vibrational energy (v) evolves within a given information-density environment**.

This perspective unifies gravitational time dilation, cosmological time evolution, and the arrow of time under a single physical mechanism: **the dependence of vibrational dynamics on the ratio** R/m.

4.1 Time as the Rate of Vibrational Evolution

In NEO-TEEM, the fundamental "clock" of a region is the rate at which its vibrational energy state v can evolve. This rate is controlled by the available spatial degrees of freedom per unit of condensed energy:

$$t_{\text{local}} \propto v(R/m).$$

Here:

- **v** sets the intrinsic oscillation or fluctuation rate,
- **R/m** determines how freely those oscillations can occur.

A region with high information density (low R/m) restricts vibrational motion, slowing the passage of time. Conversely, regions with high R/mallow rapid vibrational evolution, accelerating local time.

Thus:

Time flows faster where space is "looser" (high R/m) and slower where space is "tighter" (low R/m).

This replaces geometric curvature with a physically transparent mechanism.

4.2 Gravitational Time Dilation from Information Density

General Relativity predicts that clocks run slower in gravitational wells. In NEO-TEEM, this effect arises naturally because condensed energy reduces the surrounding information capacity R.

Let a mass Mreduce R according to:

$$R(r) = R_\infty - \frac{kM}{r}.$$

Then the local time rate becomes:

$$t(r) \propto v\left(\frac{R(r)}{m}\right) = v\left(\frac{R_\infty}{m} - \frac{kM}{mr}\right).$$

As $r$decreases:

- $R(r)$decreases,
- R/mdecreases,
- vibrational evolution slows,
- **time dilates**.

This reproduces gravitational time dilation without invoking curvature.

4.3 Cosmological Time and the Growth of R

On cosmological scales, the universe evolves through the irreversible increase of R. As R grows:

- R/mincreases globally,
- vibrational evolution accelerates,
- the "cosmic clock" speeds up.

This provides a physical interpretation of cosmological time:

Cosmic time is the cumulative effect of increasing spatial information capacity.

In this view, the expansion of the universe is not the stretching of a metric but the **growth of R**, which increases the rate at which physical processes unfold.

This explains why:

- early-universe processes occurred rapidly (low R → slow time),
- late-universe processes occur more slowly (high R → fast time),
- redshift corresponds to the historical evolution of R.

4.4 The Arrow of Time

The arrow of time emerges from the monotonic increase of R:

$$\frac{dR}{dt} > 0.$$

Because R never decreases:

- information capacity always grows,
- vibrational evolution never reverses,
- entropy increases,
- and time has a preferred direction.

This provides a microscopic foundation for thermodynamic irreversibility.

4.5 Relativistic Effects as Information-Density Effects

NEO-TEEM reproduces key relativistic phenomena:

Time dilation (special relativity)

High velocities increase effective information density by compressing available degrees of freedom, reducing $R/m$and slowing time.

Gravitational redshift

Photons lose vibrational energy when climbing out of low-R/m regions.

Shapiro delay

Light slows in regions of reduced R, analogous to optical refraction.

All these effects arise from the same mechanism: **vibrational evolution depends on information density.**

4.6 Summary

- Time is the rate of vibrational evolution.
- This rate depends on the information density R/m.
- Gravitational time dilation arises from reductions in R near mass.
- Cosmological time accelerates as R grows.
- The arrow of time follows from the monotonic increase of R.
- Relativistic effects emerge naturally from information-density variations.

This establishes time as a dynamical, emergent property of the information structure of space, completing the unification of gravity and temporal evolution within NEO-TEEM.

The next section applies this framework to cosmology, deriving an expansion history that requires no dark matter or dark energy.

5. Cosmology Without Dark Matter or Dark Energy

The standard cosmological model (ΛCDM) explains the observed expansion history of the universe by introducing two hypothetical components: **dark matter**, to enhance gravitational clustering, and **dark energy**, to drive late-time acceleration. Neither component has been directly detected, and both function as effective parameters rather than physical entities.

NEO-TEEM provides an alternative explanation in which the expansion history arises naturally from the dynamics of the three energy states (v, m, R) and the evolution of information density. In this framework:

- **cosmic acceleration** results from the growth of spatial degrees of freedom R,
- **enhanced gravitational effects** arise from the conversion of m into R,
- **radiation redshift** corresponds to the direct transfer of v into R.

This section derives the cosmological expansion equation of NEO-TEEM and shows how it reproduces observational features typically attributed to dark matter and dark energy.

5.1 Energy Components in an Expanding Universe

Each of the three energy states evolves differently with redshift:

Condensed energy (m)

Behaves like ordinary matter, diluting as

$$\rho_m(z) = \rho_{m0}(1+z)^3.$$

Vibrational energy (v)

Redshifts according to a generalized power law,

$$\rho_v(z) = \rho_{v0}(1+z)^{n_v},$$

where $n_v \approx 4$for radiation but may differ depending on the rate of v→R conversion.

Spatial degrees of freedom (R)

Increase through the conversion of m and v:

$$\rho_R(z) = \rho_{R0} + \eta[\rho_m(z) + \rho_v(z)].$$

Here:

- η quantifies the rate at which localized or vibrational energy contributes to the growth of R,
- **ρ _R0** is the present-day information capacity of space.

This formulation captures the irreversible increase of R, which drives cosmic expansion.

5.2 The NEO-TEEM Friedmann Equation

The expansion rate is determined by the total effective energy density:

$$H^2(z) = \frac{8\pi G}{3}[\rho_m(z) + \rho_v(z) + \rho_R(z)].$$

Substituting the expressions above and normalizing by H_0, we obtain:

$$H^2(z) = H_0^2[(1+\eta)\Omega_m(1+z)^3 + (1+\eta)\Omega_v(1+z)^{n_v} + \Omega_R].$$

This equation has three key features:

1. **No dark matter term** is required.
2. **No dark energy term** is required.
3. The expansion history is governed by the evolution of R.

5.3 Effective Matter Density Without Dark Matter

In ΛCDM, the matter density parameter is approximately

$$\Omega_m^{\Lambda\mathrm{CDM}} \approx 0.3,$$

even though baryonic matter contributes only

$$\Omega_b \approx 0.05.$$

NEO-TEEM explains this discrepancy through the conversion of m into R:

$$\Omega_m^{\mathrm{eff}} = (1+\eta)\Omega_m.$$

For $\eta \approx 1$and $\Omega_m \approx 0.15$:

$$\Omega_m^{\mathrm{eff}} \approx 0.30,$$

matching the effective matter density inferred from structure formation and gravitational lensing—**without invoking dark matter**.

5.4 Cosmic Acceleration Without Dark Energy

The term $\Omega_R$plays the role of dark energy, but with a clear physical meaning:

- It represents the **current value of the spatial information capacity**,
- not a cosmological constant or vacuum energy.

As R grows, its contribution becomes dominant at low redshift, producing late-time acceleration:

$$H^2(z) \approx H_0^2[\Omega_R + (1+\eta)\Omega_m(1+z)^3].$$

This reproduces the observed transition from deceleration to acceleration at $z \approx 0.6$.

Thus:

Cosmic acceleration is the dynamical consequence of increasing spatial degrees of freedom, not a mysterious negative-pressure fluid.

5.5 Early- and Late-Time Behavior

Early universe (high z)

Matter and vibrational energy dominate:

$$H(z) \propto (1+z)^{3/2}.$$

This matches the standard matter-dominated era.

Late universe (low z)

R dominates:

$$H(z) \to H_0\sqrt{\Omega_R}.$$

This mimics a cosmological constant but arises from entropy growth.

5.6 Comparison with ΛCDM

The NEO-TEEM expansion history is nearly indistinguishable from ΛCDM for appropriate parameter choices:

- $\eta \approx 1$
- $n_v \approx 4$
- $\Omega_m \approx 0.15$
- $\Omega_R \approx 0.65$

Yet the physical interpretation is entirely different:

ΛCDM	NEO-TEEM
Dark matter required	Effective mass from m→R conversion
Dark energy required	R-growth drives acceleration
Geometry is fundamental	Information density is fundamental
Expansion is metric stretching	Expansion is increase of R

NEO-TEEM therefore provides a **conceptually simpler and physically grounded** cosmology.

5.7 Summary

- The expansion history of the universe arises from the dynamics of v, m, and R.
- No dark matter or dark energy is required.
- Effective matter density increases through m→R conversion.
- Cosmic acceleration results from the growth of R.
- The NEO-TEEM Friedmann equation reproduces ΛCDM-like behavior with clear physical

meaning.

The next section tests this expansion history against Type Ia supernova observations.

7. Black Holes as Information-Density Structures (Overview)

Black holes represent the most extreme environments in the universe, where conventional descriptions of spacetime, matter, and information break down. In General Relativity, black holes are geometric objects characterized by event horizons and singularities. In NEO-TEEM, they are instead understood as **regions of extreme information density**, where the ratio R/mapproaches its minimum physically attainable value. This reinterpretation eliminates the need for singularities, clarifies the origin of black-hole entropy, and provides a unified explanation for horizon behavior, Hawking-like radiation, and gravitational lensing.

7.1 Horizons as Steep Information-Density Gradients

In NEO-TEEM, the event horizon is not a geometric boundary but a **transition layer** where the information density R/mchanges extremely rapidly.

Let the information density near a compact object behave as:

$$\frac{R(r)}{m(r)} \to I_{\min} \text{as } r \to r_H.$$

Key features:

- $I_{\min}$is finite, so **no singularity** arises.
- The horizon corresponds to the radius where **vibrational evolution freezes**,

$$v(R/m) \to 0,$$

producing the familiar "time stops at the horizon" effect.

- The steep gradient $\nabla(R/m)$generates the strong gravitational field.

Thus:

A black hole is a region where information density reaches its minimum allowable value, not a point of infinite curvature.

7.2 Entropy as Boundary Information Capacity

The Bekenstein–Hawking entropy formula suggests that black-hole entropy scales with horizon area. In NEO-TEEM, this arises naturally because:

- the horizon is a **boundary layer of constrained R**,
- the number of accessible microstates is proportional to the number of boundary degrees of freedom,
- the interior contributes little because $R/m$is nearly constant and minimal.

Thus, the entropy is:

$$S \propto R_{\text{boundary}} \propto A_H,$$

where $A_H$is the horizon area.

This provides a **physical** (not geometric) explanation for the area law.

7.3 Avoidance of Singularities

Because R cannot decrease below a finite minimum and m cannot increase without bound, the ratio R/mnever reaches zero. Therefore:

- no infinite curvature,
- no infinite density,
- no breakdown of physical law.

Instead, the interior of a black hole is a **finite-information-density core**, where:

- time evolution is extremely slow,
- vibrational modes are suppressed,
- but the structure remains physically meaningful.

This resolves the classical singularity problem.

7.4 Hawking-Like Radiation as R-Reconfiguration

In NEO-TEEM, Hawking radiation is not a quantum-pair-creation process at the horizon. Instead, it arises from **slow reconfiguration of R** in the boundary layer:

- fluctuations in R allow small amounts of vibrational energy to escape,
- the spectrum is approximately thermal but not exactly Planckian,
- information is preserved because R evolves continuously.

Thus:

Black-hole evaporation is a slow leakage of vibrational energy driven by changes in spatial information capacity.

This provides a natural resolution to the information-loss paradox.

7.5 Gravitational Lensing as Spatial Refraction

Because the horizon region has extremely low R/m, it acts as a **high-index medium** for light:

$$n(r) \approx \frac{R_{\mathrm{vac}}}{R(r)}.$$

As $R(r)$decreases near the horizon, $n(r)$increases sharply, bending light in a manner identical to optical refraction.

This yields:

- the photon sphere,
- the characteristic EHT shadow,
- strong lensing arcs,
- and the brightness asymmetry of accretion flows.

Thus:

Gravitational lensing is the optical consequence of spatial information-density gradients.

7.6 Connection to Cosmic Information Dynamics

Black holes are not isolated anomalies but **extreme manifestations of the same R-dynamics** that govern cosmic expansion.

- In cosmology, R increases globally.
- In black holes, R decreases locally to its minimum.
- Both phenomena arise from the same energy-flow hierarchy $v \to m \to R$.

This unifies black-hole physics with cosmology under a single principle.

7.7 Summary

- Black holes are finite information-density structures, not geometric singularities.
- Horizons correspond to steep gradients in R/m.
- Entropy arises from boundary information capacity.
- Hawking-like radiation is driven by R-reconfiguration.
- Gravitational lensing is spatial refraction in low-R regions.
- Black holes and cosmic expansion are two limits of the same information-dynamic process.

A full treatment of these ideas—including interior structure, horizon thickness, shadow morphology, and evaporation dynamics—will be presented in the companion paper *NEO-TEEM: Black Holes and Lensing*.

8. Refraction, Lensing, and Information Confinement (Overview)

Light propagation is traditionally described by two separate mechanisms:

- **optical refraction**, arising from the electromagnetic response of matter, and
- **gravitational lensing**, arising from spacetime curvature in General Relativity.

NEO-TEEM unifies these phenomena under a single principle: **light bends wherever the information density R/mchanges.**

In this framework, both optical materials and gravitational fields act as **information-confining media** that modify the effective refractive index of space. This section outlines the unified description and its implications.

8.1 Refractive Index as Information Confinement

In NEO-TEEM, the refractive index of a material is not a property of electromagnetic polarization alone. Instead, it reflects how strongly the material **reduces the local spatial information capacity R**.

Let R_{vac}be the information capacity of vacuum and R_{med}that of a material. Then the refractive index is approximately:

$$n \approx \frac{R_{\text{vac}}}{R_{\text{med}}}.$$

Thus:

- materials with **lower R** have **higher n**,

- light slows because vibrational modes propagate less freely in low-R environments,
- optical refraction is the direct consequence of **information confinement**.

This interpretation connects optical properties to the same physical quantity that governs gravity.

8.2 Spatial Refraction from Information-Density Gradients

Light follows the path that maximizes the local vibrational evolution rate. Since this rate depends on R/m, the trajectory of light is determined by:

$$\text{light path} = \text{extremal curve of } v(R/m).$$

Taking the variation yields a generalized Snell-like law:

$$n(r)\sin\,\theta = \text{constant},$$

where

$$n(r) = \frac{R_{\text{vac}}}{R(r)}.$$

This applies equally to:

- optical interfaces,
- gravitational fields,
- black-hole environments,
- and any region with spatially varying R.

Thus:

Refraction is the universal response of light to information-density gradients.

8.3 Gravitational Lensing as a Form of Refraction

In a gravitational field, the presence of mass reduces R according to:

$$R(r) = R_\infty - \frac{kM}{r}.$$

The corresponding refractive index is:

$$n_{\text{grav}}(r) = \frac{R_{\text{vac}}}{R(r)}.$$

As $r$decreases:

- R decreases,
- n_{grav}increases,
- light bends toward the mass.

This reproduces:

- weak lensing,
- strong lensing,
- Einstein rings,
- photon spheres,
- and the EHT black-hole shadow.

All arise from the same mechanism as optical refraction.

8.4 Unified Interpretation of Optical and Gravitational Bending

The NEO-TEEM framework yields a single equation for light bending:

$$\frac{d^2\mathbf{x}}{ds^2} = -\nabla \ln\ n(r) = \nabla \ln\ R(r).$$

Thus:

- **optical refraction**: ∇Rarises from atomic structure.
- **gravitational lensing**: ∇Rarises from mass-induced compression of spatial degrees of freedom.
- **black-hole lensing**: ∇Rbecomes extremely steep near the horizon.

This unifies all known bending phenomena under a single physical principle.

8.5 Material Properties as Probes of R

Because refractive index directly measures R-confinement, ordinary materials provide experimental access to the information structure of matter.

Examples:

- Water: $n = 1.33$→ moderate R-confinement
- Glass: $n = 1.5$→ stronger confinement
- Diamond: $n = 2.42$→ very strong confinement

These values correspond to:

$$\Delta R = R_{\text{vac}} - R_{\text{med}} \propto n - 1.$$

Thus:

Optical materials encode the information-density structure of condensed matter.

This connection will be developed in detail in the companion paper on lensing and optical behavior.

8.6 Implications for Black-Hole Optics

Near a black-hole horizon:

- $R(r)$approaches its minimum value,
- $n(r)$becomes extremely large,
- light experiences extreme spatial refraction.

This yields:

- the sharp boundary of the EHT shadow,
- the brightness asymmetry of accretion flows,
- the photon ring structure,
- and the suppression of radial propagation.

These features arise without invoking spacetime curvature— they follow directly from the information-density profile.

8.7 Summary

- Refractive index measures the degree of information confinement.
- Optical refraction and gravitational lensing arise from the same mechanism: **gradients in R**.
- Light follows extremal paths of vibrational evolution determined by R/m.
- Black-hole lensing is extreme spatial refraction in low-R regions.
- Material refractive indices provide experimental probes of R-structure.

A full treatment—including quantitative models of refractive profiles, lensing equations, and black-hole shadow morphology—will be presented in the companion paper *NEO-TEEM: Black Holes and Lensing*.

9. Discussion

NEO-TEEM proposes a unified physical framework in which gravity, time, entropy, and cosmic expansion emerge from the dynamics of three energy states—vibrational energy (v), condensed energy (m), and spatial degrees of freedom (R). The central organizing principle is the **information density** R/m, whose gradients govern gravitational behavior and whose evolution drives the expansion of the universe.
This section discusses the broader implications of the framework, its relationship to existing theories, its predictive power, and the open questions that remain.

9.1 Conceptual Unification

NEO-TEEM replaces several independent concepts in modern physics with a single underlying mechanism:

- **Gravity** arises from gradients in information density.
- **Time** is the rate of vibrational evolution controlled by R/m.
- **Entropy** is the growth of R.
- **Cosmic expansion** is the global increase of R.
- **Black holes** are regions where R/mreaches its minimum.
- **Refraction and lensing** are responses to spatial variations in R.

This unification eliminates the need for:

- geometric curvature as a fundamental entity,
- dark matter as an additional mass component,
- dark energy as a negative-pressure fluid,
- singularities inside black holes,
- separate optical and gravitational bending mechanisms.

Instead, all these phenomena follow from the same physical quantity: **the distribution and evolution of R**.

9.2 Relation to General Relativity and Λ CDM

NEO-TEEM is not a modification of General Relativity but a **reinterpretation of its phenomenology**.

- GR's curvature corresponds to reductions in R.
- Gravitational time dilation arises from suppressed vibrational evolution.

- Geodesics correspond to extremal paths of $v(R/m)$.
- The cosmological constant corresponds to the present value of R.
- Dark matter corresponds to effective mass enhancement from m→R conversion.

Thus, GR and ΛCDM emerge as **effective descriptions** of deeper information-dynamic processes.
This perspective explains why GR works so well in weak fields while offering new insights into strong-field and cosmological regimes.

9.3 Predictive Power and Observational Tests

NEO-TEEM makes several predictions that distinguish it from standard cosmology and gravity:

1. Non-Planckian deviations in black-hole radiation

Hawking-like emission should show small but measurable departures from a perfect thermal spectrum due to R-reconfiguration dynamics.

2. Horizon thickness and shadow structure

The EHT shadow should exhibit subtle edge-broadening effects corresponding to the finite gradient scale of R.

3. Modified lensing profiles

Strong lensing near compact objects should follow refractive-index-like behavior rather than purely geometric deflection.

4. Effective matter density scaling

Galaxy rotation curves and cluster dynamics should follow

$$\Omega_m^{\mathrm{eff}} = (1+\eta)\Omega_m,$$

with η measurable from cosmological data.

5. Redshift–time relation

The cosmic time evolution inferred from high-z objects should reflect the growth of R rather than metric expansion.
These predictions provide clear avenues for empirical validation.

9.4 Advantages Over Existing Frameworks

NEO-TEEM offers several conceptual and practical advantages:

- **No exotic components** (dark matter, dark energy) are required.
- **No singularities** arise in black-hole interiors.
- **Time, entropy, and gravity** share a common physical origin.
- **Light propagation** is unified across optical and gravitational contexts.
- **Cosmic acceleration** emerges naturally from R-growth.

- **The theory is minimal**, relying on a single ratio R/mrather than multiple fields or potentials.

This simplicity is a major strength, offering a coherent physical picture without sacrificing empirical accuracy.

9.5 Open Questions and Future Directions

Despite its unifying power, several questions remain open:

1. Microphysical origin of R

What is the fundamental structure that gives rise to spatial degrees of freedom?

2. Quantum behavior

How does NEO-TEEM connect to quantum measurement, entanglement, and decoherence?

3. Early-universe initial conditions

What determines the initial distribution of v, m, and R?

4. Horizon microstructure

What is the detailed behavior of R near the minimum information-density limit?

5. Nonlinear dynamics of m→R conversion

How does the conversion rate η vary across cosmic epochs and environments?

These questions define the research program for future work, including the companion papers on cosmology and black-hole physics.

9.6 Summary

NEO-TEEM provides a unified, physically transparent framework in which:

- gravity,
- time,
- entropy,
- cosmic expansion,
- black-hole behavior,
- and light propagation

all arise from the dynamics of information density.

The theory reproduces the successes of GR and ΛCDM while offering new insights and testable predictions. It replaces geometric curvature with a more fundamental quantity—**the distribution and evolution of spatial information capacity**—and thereby offers a coherent path toward a deeper understanding of the universe.

10. Conclusion

This work has introduced **NEO-TEEM**, a unified physical framework in which gravity, time, entropy, cosmic expansion, black-hole behavior, and light propagation all emerge from the dynamics of three fundamental energy states: vibrational energy (v), condensed energy (m), and spatial degrees of freedom (R). The central organizing principle of the theory is the **information density** R/m, whose gradients generate gravitational attraction and whose evolution governs the temporal and cosmological behavior of the universe.

We have shown that:

- **Gravity** arises from spatial variations in information density,

$$g = -\nabla\left(\frac{R}{m}\right),$$

providing a physically transparent alternative to geometric curvature.

- **Time** is the local rate of vibrational evolution, controlled by the value of R/m, yielding natural explanations for gravitational and relativistic time dilation.
- **Entropy** corresponds to the growth of R, linking thermodynamics to cosmic evolution.
- **Cosmic expansion** emerges from the irreversible increase of R, eliminating the need for dark energy.
- **Effective gravitational mass** increases through the conversion $m \to R$, reproducing the observational role of dark matter without invoking new particles.
- **Black holes** are finite information-density structures with no singularities, where horizons correspond to steep gradients in $R/m$and Hawking-like radiation arises from slow reconfiguration of R.
- **Refraction and gravitational lensing** are unified as manifestations of spatial information-density gradients, with optical materials and gravitational fields described by the same refractive-index formalism.

Together, these results demonstrate that a wide range of physical phenomena—traditionally treated as separate domains—can be understood as different expressions of a single underlying mechanism: **the distribution and evolution of spatial information capacity**.

NEO-TEEM reproduces the empirical successes of General Relativity and ΛCDM while offering a simpler conceptual foundation and new, testable predictions. It provides a coherent path toward resolving long-standing issues such as the nature of dark components, the origin of cosmic acceleration, the meaning of entropy, and the internal structure of black holes.

Two companion papers will extend this work:

- **NEO-TEEM Cosmology**, presenting detailed fits to supernova, H(z), and CMB data;
- **NEO-TEEM Black Holes and Lensing**, developing the full information-density model of horizons, shadows, and refractive light propagation.

The results presented here suggest that information density may be the fundamental quantity underlying the structure and evolution of the universe. If so, NEO-TEEM offers a unified and physically grounded framework for understanding the cosmos at all scales—from the smallest vibrational modes to the largest cosmic structures.

Detailed observational comparisons, numerical simulations, and graphical analyses will be presented in the companion papers on cosmology and black-hole physics.